

An Electronics KALEIDOSCOPE

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Attention-catching elements prove valuable in warnings, raising awareness of danger, and in exhibitions, providing visual delight. Numerous electronic circuits utilise blinkers, winkers, and LED flashers for these purposes.

This mosaic display of LEDs, reminiscent of a kaleidoscope, exudes charm and fascination. As this is a straightforward detection system, you have the freedom to construct and present it any way

you like for amusement. The author's prototype is shown in Fig. 1.

Circuit and working

Fig. 2 shows circuit diagram of an electronics kaleidoscope. The circuit comprises a 555 timer (IC1), a 14-bit binary counter/divider 4020 (IC2), and nine different-coloured 5mm LEDs (red, orange, pink, green, yellow, blue, white, etc) with LED holders (LED1 through LED9) and various other components. A 9V power adaptor may be used for the power supply.

The 555 timer is configured as an astable multivibrator. It connects output pin 3 directly to timing capacitor C1 via resistor R1. When the output at pin 3 is high, capacitor C1 charges through resistor R1. When the voltage across the capacitor reaches $2/3V_{CC}$, pin 6 triggers the output at pin 3 to shift to a low state.

The capacitor then discharges back through

the same resistor until pin 2 reaches $1/3V_{CC}$, causing the output to change state once again. The capacitor continuously charges and discharges between $2/3V_{CC}$ and $1/3V_{CC}$, creating a high and low state at the output pin 3. Here, V_{CC} is 9V.

As the capacitor charges and discharges through the same resistor, the duty cycle of this basic

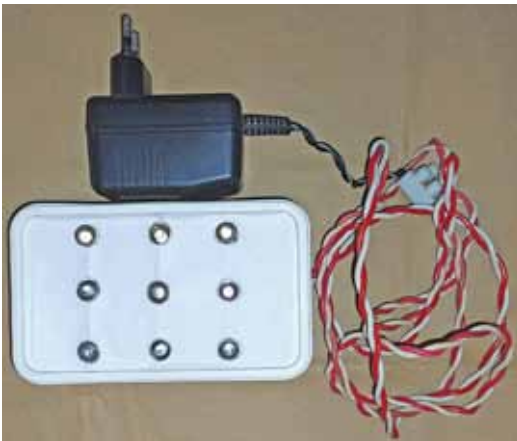


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

- IC1 - 555 timer
- IC2 - 4020 14-bit binary counter/divider
- LED1-LED9 - 9 different coloured 5mm LEDs with holders

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 33-kilo-ohm
- R2-R10 - 1-kilo-ohm

Capacitors:

- C1 - 10 μ F, 25V electrolytic
- C2 - 0.1 μ F polyester

Miscellaneous:

- Thin enamelled copper wire
- 8-pin IC socket (optional)
- 16-pin IC socket (optional)
- Small flat plastic box
- Flexible wire
- 9V DC adaptor

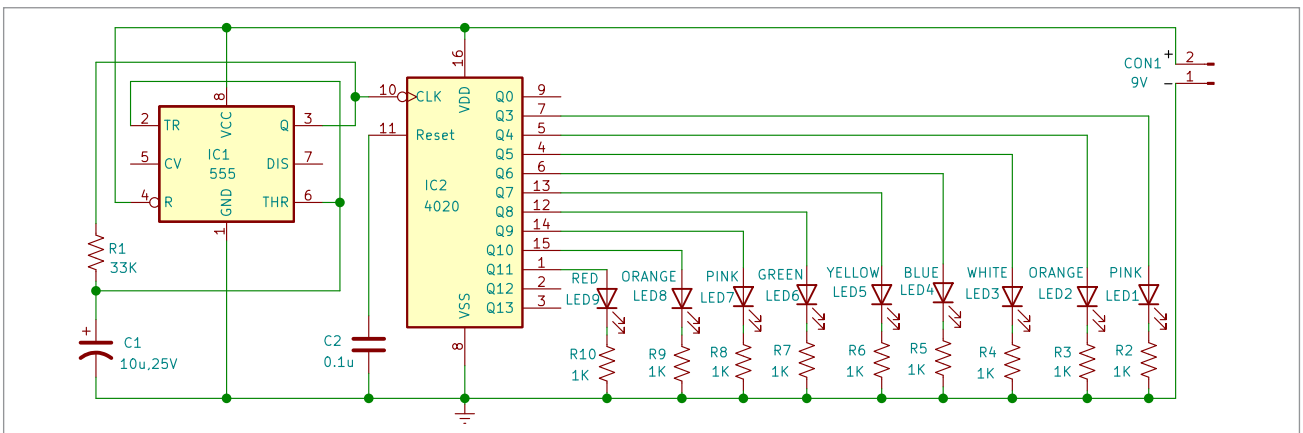


Fig. 2: Circuit diagram